This assignment will not be graded and is only for practice.

Level 1

1) Consider a function $f : \mathbb{R}^3 \to \mathbb{R}$. **1 point**

Choose the correct statement with respect to ∇f.

- [ ] ∇f has co-domain $\mathbb{R}^2$
- [ ] ∇f has co-domain $\mathbb{R}$
- [ ] ∇f has co-domain $\mathbb{R}^3$
- [ ] ∇f has domain $\mathbb{R}^3$

No, the answer is incorrect.
Score: 0

Accepted Answers:

∇f has co-domain $\mathbb{R}^3$
∇f has domain $\mathbb{R}^3$

2) Let $f : \mathbb{R}^3 \to \mathbb{R}$ and suppose its gradient ∇f is continuous. Which of the following statements is true for the directional derivative of f in the direction of $a \in \mathbb{R}^3$? Consider dot product to be the inner product $<,>$. **1 point**

- ( ) $D_a f = \frac{1}{\|a\|} < a, \nabla f >$
- ( ) $D_a f =< a, \nabla f >$
- ( ) $D_a f = \frac{1}{2\|a\|} < a, \nabla f >$
- ( ) $D_a f = \frac{1}{\|a\|^2} < a, \nabla f >$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$D_a f = \frac{1}{\|a\|} < a, \nabla f >$

3) Let $f(x, y) = e^x \sin y$. If ∇f at the point $(ln(2), \pi/4)$ is (k, k) where $k \in \mathbb{R}$, then find the value of k^2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Level 2

4) Let $p = -\frac{k}{(\|r\|)^3} r$, where $r = (x, y, z) \in \mathbb{R}^3 \setminus \{(0,0,0)\}, k \in \mathbb{R}$. **_1 point_**

Amongst the options below, choose those for which the function f satisfies $\nabla f = p$ for some $k \in \mathbb{R}$.

Consider dot product as the inner product for finding the norm.

- [ ] $f(x, y, z) = \frac{1}{\|r\|}$
- [ ] $f(x, y, z) = \frac{1}{2\|r\|}$
- [ ] $f(x, y, z) = \frac{1}{\|r\|+2}$
- [ ] $f(x, y, z) = \|r\|$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$f(x, y, z) = \frac{1}{\|r\|}$
$f(x, y, z) = \frac{1}{2\|r\|}$

5) Let $f(x, y, z) = 4(x^2 + y^2) - z^2$. **_1 point_**

Choose the correct option(s).

- [ ] ∇f at $(1, 0, 2)$ is $(8, 0, -4)$
- [ ] ∇f is continuous everywhere in $\mathbb{R}^3$
- [ ] ∇f is not continuous everywhere in $\mathbb{R}^3$
- [ ] ∇f at $(1, 0, 2)$ is $(-8, 0, -4)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

∇f at $(1, 0, 2)$ is $(8, 0, -4)$
∇f is continuous everywhere in R^3

6) Consider a function defined as **1 point**

$$f(x, y) = \begin{cases} \frac{y^3}{x^2+y^2} & (x, y) = (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}.$$

Choose the correct option(s).

- [ ] $\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2+y^2)^2}$, for $(x, y) = (0, 0)$
- [ ] $\frac{\partial f}{\partial x} = \frac{2xy^3}{(x^2+y^2)^2}$, for $(x, y) = (0, 0)$
- [ ] $\frac{\partial f}{\partial y} = \frac{3x^2y^2+y^4}{(x^2+y^2)^2}$ for $(x, y) = (0, 0)$
- [ ] $\frac{\partial f}{\partial y} = \frac{3x^2y^2-y^4}{(x^2+y^2)^2}$ for $(x, y) = (0, 0)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2+y^2)^2}$, for $(x, y) = (0, 0)$
$\frac{\partial f}{\partial y} = \frac{3x^2y^2+y^4}{(x^2+y^2)^2}$ for $(x, y) = (0, 0)$

7) Consider the function definition given in Q6 above. Choose the correct option(s). **1 point**

- [ ] $\frac{\partial f}{\partial x}$ is continuous at $(0, 0)$
- [ ] $\frac{\partial f}{\partial x}$ is not continuous at $(0, 0)$
- [ ] $\frac{\partial f}{\partial y}$ is not continuous at $(0, 0)$
- [ ] ∇f is discontinuous at the origin $(0, 0)$
- [ ] Directional derivative of f in the direction of a vector $a \in \mathrm{R}^2$ at $(0, 0)$ can be obtained using the gradient of f

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\frac{\partial f}{\partial x}$ is not continuous at $(0, 0)$
$\frac{\partial f}{\partial y}$ is not continuous at $(0, 0)$
∇f is discontinuous at the origin $(0, 0)$

8) Consider a function defined as **1 point**

$$f(x, y) = \begin{cases} (x^2 + y^2)\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) & (x, y) = (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}.$$

Choose the correct option(s).

- [ ] $\frac{\partial f}{\partial y}(0, 0) = 0$
- [ ] $\frac{\partial f}{\partial x}(0, 0) = 0$
- [ ] $\frac{\partial f}{\partial x} = 0$
- [ ] $\frac{\partial f}{\partial y} = 0$
- [ ] None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\frac{\partial f}{\partial y}(0, 0) = 0$
$\frac{\partial f}{\partial x}(0, 0) = 0$

9) Consider the function definition given in Q8 above. Choose the correct option(s) with respect to the partial derivative of f. Note $sign(x) = \pm 1$ depending on the sign of x. **1 point**

- [ ] $\frac{\partial f}{\partial x}(x, 0) = 2x\sin(\frac{1}{|x|}) - sign(x)\cos(\frac{1}{|x|})$ for $x = 0$
- [ ] $\frac{\partial f}{\partial x}(x, 0) = 2x\sin(\frac{1}{|x|}) - sign(x)\cos(\frac{1}{|x|})$

- [ ] $\frac{\partial f}{\partial x}(x,0) = 2x\sin(\frac{1}{|x|}) - \cos(\frac{1}{|x|})$ for $x = 0$
- [ ] $\frac{\partial f}{\partial y}(0,y) = 2y\sin(\frac{1}{|y|}) - sign(y)\cos(\frac{1}{|y|})$ for $y = 0$
- [ ] $\frac{\partial f}{\partial y}(0,y) = 2y\sin(\frac{1}{|y|}) - sign(y)\cos(\frac{1}{|y|})$
- [ ] $\frac{\partial f}{\partial y}(0,y) = 2y\sin(\frac{1}{|y|}) - \cos(\frac{1}{|y|})$ for $y = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial f}{\partial x}(x,0) = 2x\sin(\frac{1}{|x|}) - sign(x)\cos(\frac{1}{|x|})$ for $x = 0$
$\frac{\partial f}{\partial y}(0,y) = 2y\sin(\frac{1}{|y|}) - sign(y)\cos(\frac{1}{|y|})$ for $y = 0$

10) Consider the function definition given in Q8 above. Choose the correct option. ***1 point***

- [ ] ∇f is continuous everywhere
- [ ] ∇f is discontinuous at the origin $(0,0)$
- [ ] $\frac{\partial f}{\partial x}$ is continuous at the origin but $\frac{\partial f}{\partial y}$ is discontinuous at the origin
- [ ] $\frac{\partial f}{\partial y}$ is continuous at the origin but $\frac{\partial f}{\partial x}$ is discontinuous at the origin
- [ ] Directional derivative of f in the direction of a vector $a \in \mathrm{R}^2$ at $(0,0)$ cannot be obtained using the gradient of f

No, the answer is incorrect.

Score: 0

Accepted Answers:

∇f is discontinuous at the origin $(0,0)$
Directional derivative of f in the direction of a vector $a \in \mathrm{R}^2$ at $(0,0)$ cannot be obtained using the gradient of f

Your score is: 0/10